Barcodes Print

User Manual

Direct-to-printer barcode label printing for ERPNext, via QZ Tray or Zebra Browser Print.

This manual walks through installation, configuration, day-to-day printing, and every setting available in the app.

1. Overview

Barcodes Print lets you print barcode labels directly to a local thermal printer, without a print preview, a browser print dialog, or a PDF step. Labels are sent as raw printer commands (ZPL), so barcodes print using the printer's own barcode engine rather than as a slower, softer image render.

You can print from a standalone Print Barcode page, or directly from a submitted Purchase Order, Purchase Receipt, or Subcontracting Order.

2. Installation

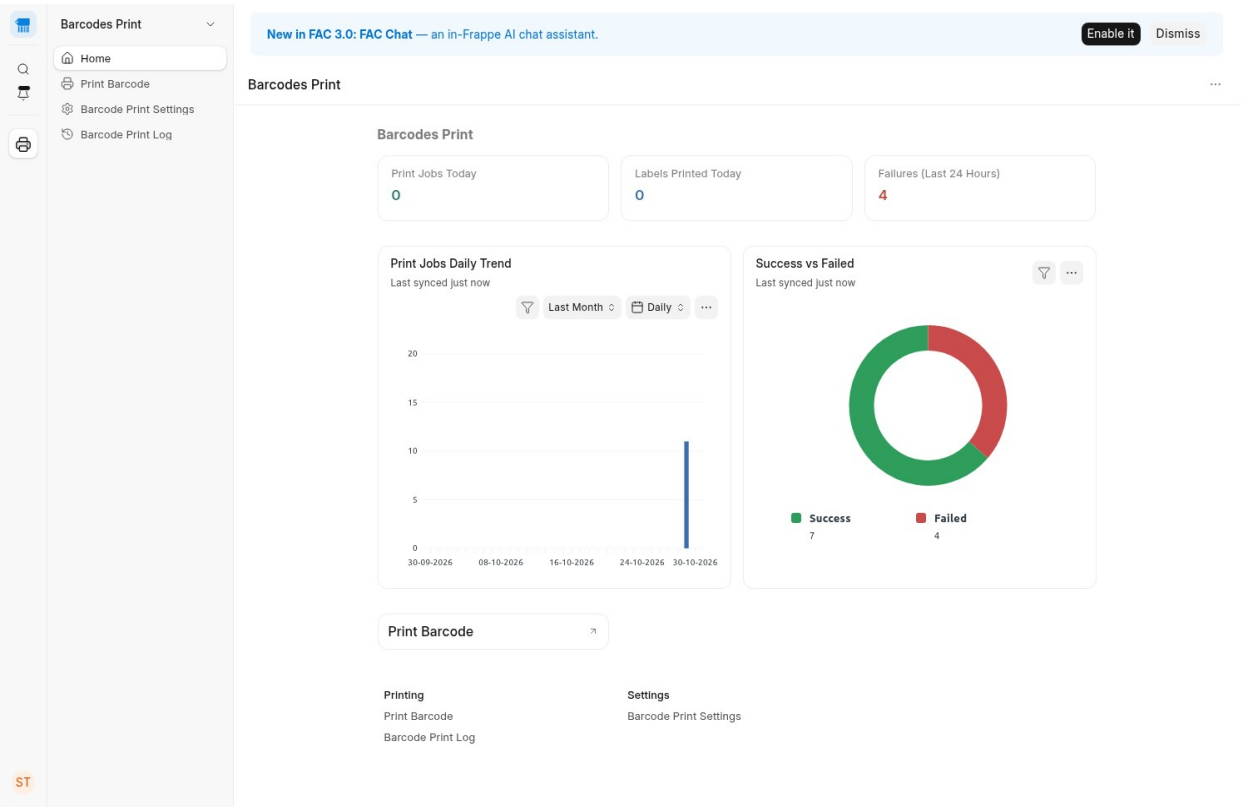
Install the app on your bench like any other Frappe app:

```
bench get-app https://github.com/Rahul-ai1/ERPNext-Barcodes-Print.git
bench --site <your-site> install-app barcodes_print
bench --site <your-site> migrate
bench restart
```

> **Note:** Installs on both Frappe v15 and v16. On v15, the Desk sidebar entry and app icon are skipped automatically (they depend on v16-only doctypes); every printing feature works the same on both versions.

3. The Barcodes Print Workspace

Once installed, Barcodes Print adds its own entry to the Desk sidebar and a home screen icon, giving you one place to reach the Print Barcode page, Settings, and the Print Log.



4. QZ Tray Setup (one time, per printer PC)

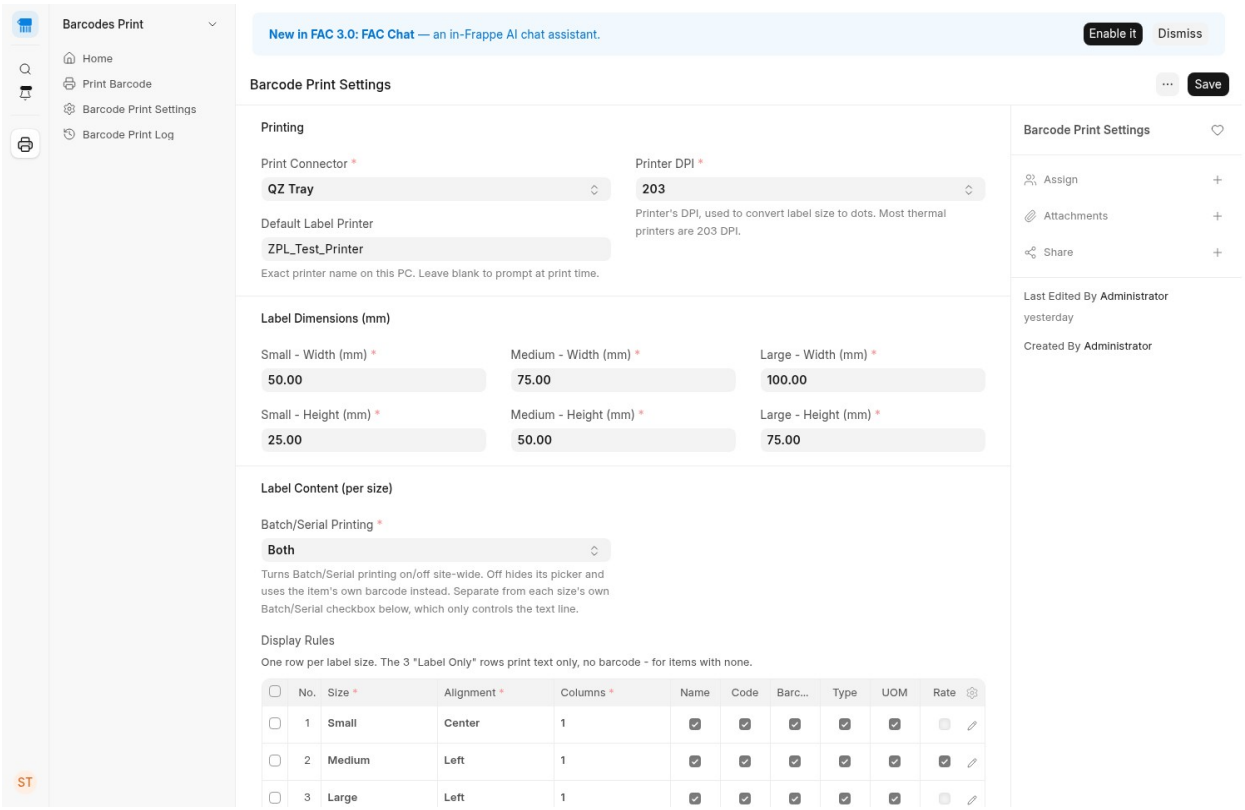
Each computer that will actually print a label needs QZ Tray running locally. This is a one-time setup per machine:

1. Download QZ Tray from qz.io/download and install it.
2. Launch it and confirm it is running (system tray on Windows, menu bar on macOS/Linux).
3. Connect the label printer to that PC and note its exact name as it appears in the operating system's printer list.
4. Set the printer's label roll size in its own OS driver preferences.
5. In Barcode Print Settings, set the exact printer name under Default Label Printer (optional — if left blank, QZ Tray uses that PC's own default printer).
6. The first time a print happens from a given browser, QZ Tray shows a one-time "Action Required" popup — tick Remember this decision and click Allow.

From then on, clicking Print sends straight to the printer with no further prompts and no browser dialog.

5. Barcode Print Settings

One screen controls every aspect of printing. Open it from the Barcodes Print workspace.



5.1 Print Connector

Choose QZ Tray or Zebra Browser Print (one at a time), plus the default printer name and printer DPI.

5.2 Label Dimensions

Set the width and height, in millimetres, for each of the three label sizes: Small, Medium, and Large.

5.3 Batch/Serial Printing

One site-wide switch decides whether printing by Batch No. or Serial No. is available at all: Both, Batch Only, Serial Only, or Neither. When a type is switched off here, its picker simply does not appear on the Print Barcode page, and printing always falls back to the item's own barcode instead.

> **Note:** This is separate from whether the resolved Batch No./Serial No. is shown as text once printed — that is controlled per label size in Display Rules below. A company can, for example, allow encoding Batch No. into every label's barcode here, while only showing the Batch No. text on Medium and Large labels in Display Rules.

5.4 Display Rules

A table with one row per label size, each with its own independent show/hide toggles and alignment — a field can be shown on Medium and Large but hidden on Small, for example. Each checkbox's column header is a short, plain label (Name, Code, Barcode, Type, UOM, Rate, Batch, Serial) since the row's own Size already gives the context:

Field	What it controls
Name	Show the item's name on the label
Code	Show the item code on the label
Barcode	Show the barcode value as text under the bars

Type	Show the barcode symbology (e.g. EAN-13, Code 128)
UOM	Show the unit of measure
Rate	Show the price (see Pricing below)
Batch / Serial	Show the resolved batch or serial number as text on the label — only relevant when Batch/Serial Printing above allows that type
Alignment	Left, Center, or Right — applies to the whole label's content (or to each column separately, if Columns is 2 — see below)
Columns	1 (default) or 2 — see below

The table has 6 rows: Small, Medium, Large, and their three "Label Only" variants (Small/Medium/Large Label Only). A Label Only row has no Barcode/Type/Batch/Serial toggles that matter - those sizes never print a barcode at all (see section 6.5).

Columns set to 2 splits that size's enabled text fields side-by-side into two columns, printed below the barcode (the barcode itself always stays full-width above them, never shrunk or split). Fields are divided in the same order they already print in — the first half goes in column 1, the rest in column 2 — and Alignment applies within each column separately, not across the whole label's width. Saving is blocked if a size's own Label Dimensions width is under 40mm, since two legible columns need the room; the error names the row and its actual width.

5.5 Pricing

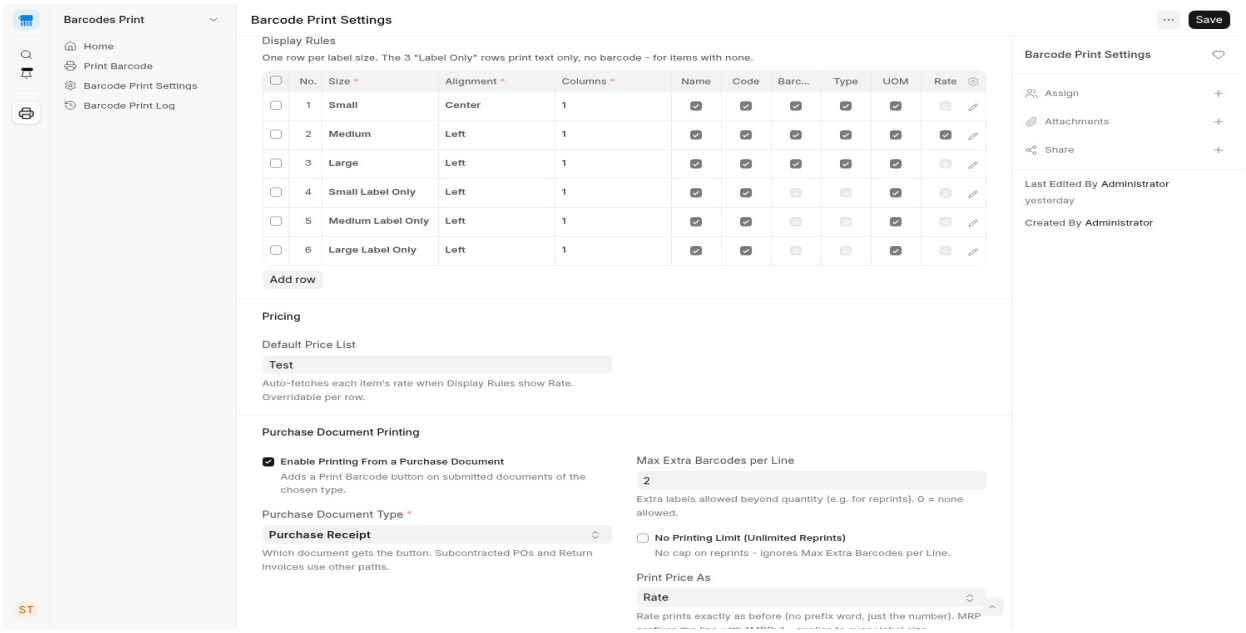
A Default Price List can be set so the Rate field on the Print Barcode page pre-fills automatically; it can still be changed per row. The Rate is always looked up from that Price List's own Item Price record, never from the Item's Standard Selling Rate — if no price is found there, the Rate line is left off the label entirely rather than printing a misleading zero or "N/A".

To print tax-inclusive pricing, pick a Company and a Tax Template on the row (both default from your own default Company, or are pulled straight from the Purchase Order/Purchase Receipt line when printing from one — see section 7). With a Tax Template set, the Rate is marked up by that template's own tax percentage and the label prints a short "Incl. of taxes" note beside it; leave Tax Template blank and the plain Price List rate prints, marked "Excl. of taxes" instead. Neither the tax percentage nor the Company/template name is ever printed — only the final Rate and that one short note.

Print Price As is a site-wide choice between "Rate" (the default - no prefix word, exactly as described above) and "MRP", which prefixes the same line with "MRP: " (e.g. "MRP: 45.00 (Excl. of taxes)"). This applies to every label size and does not require a Tax Template to be set - it simply follows whichever wording is chosen in Settings.

The Rate is also prefixed with the resolved Price List's own currency - e.g. "$2,500.00" for a USD Price List, never the site's own default currency. A real thermal printer's built-in font cannot print every currency symbol: "$" prints fine, but "₹", "€", "£", and "¥" all print as blank space on a real Zebra printer. When a currency's own symbol is not plain text the printer can render, its 3-letter code is printed instead (e.g. "INR 2,500.00") so the label is always legible.

5.6 Purchase Document Printing



Turned off by default. When enabled, pick exactly one document type — Purchase Order, Purchase Receipt, or Purchase Invoice — and a Print Barcode button appears on submitted documents of that type.

- Max Extra Barcodes per Line — how many labels beyond the line's own quantity may be printed, for damaged-label reprints. Only applies when the user raises the quantity above what the line allows; printing the line's own quantity never needs this allowance.
- Unlimited Reprints — when switched on, the per-line cap above is bypassed entirely and the same document can be reprinted any number of times. The initial pre-fill still always defaults to the line's exact quantity, never more, unless the user deliberately raises it.
- Subcontracting Orders are always supported alongside whichever of the three document types is selected, for the subcontracted-purchase-order case described below.

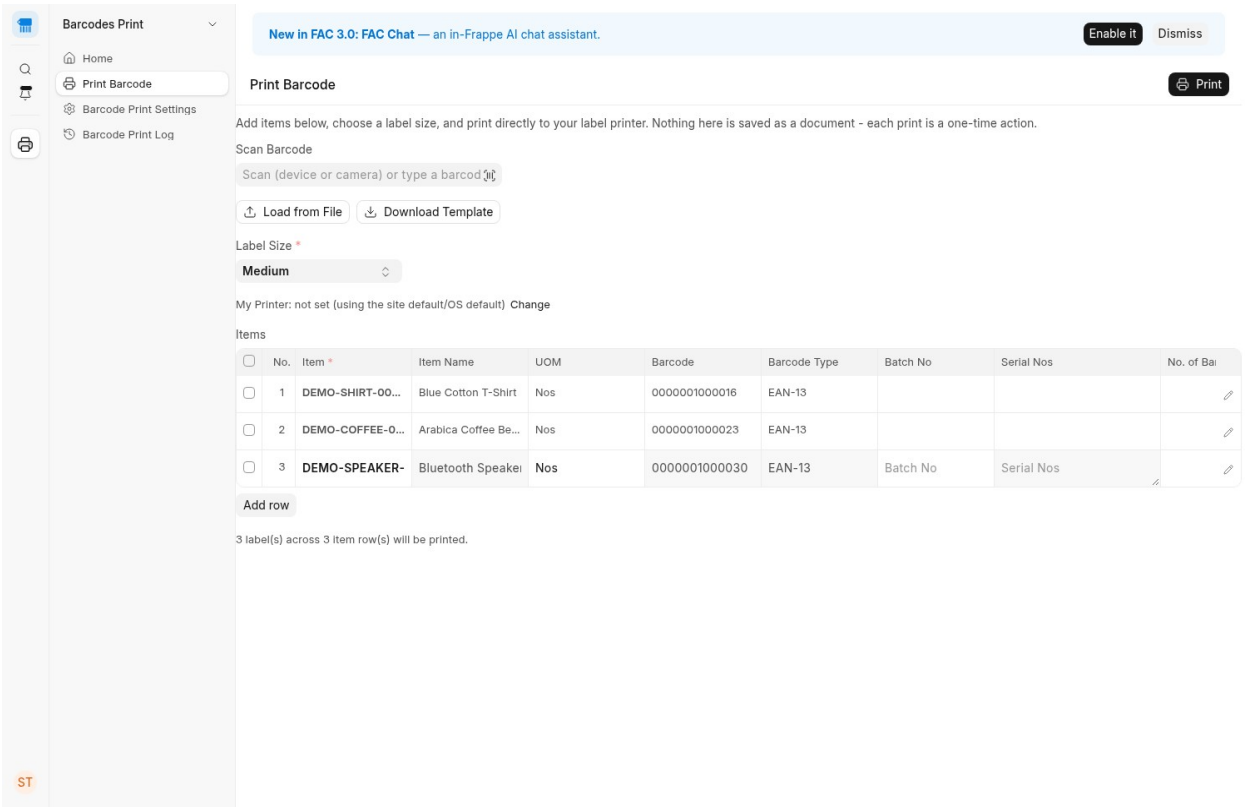
Note: Service (non-stock) item rows are automatically left out of every pre-fill — only physical, stock-tracked items get a label.

Note: A subcontracted Purchase Order does not offer barcode printing itself; use its linked Subcontracting Order instead, which prints only the finished goods.

Note: A Return (credit note) Purchase Invoice never gets the Print Barcode button - only standard submitted invoices do.

6. Printing from the Print Barcode Page

This is the main way to print. Add one row per item, pick a label size, and click Print.



1. Pick an Item in the first column — Item Name, UOM, Barcode, and Barcode Type fetch automatically.
2. Set No. of Barcodes for how many labels that row should produce.
3. Optionally pick a Price List, and a Company/Tax Template for tax-inclusive pricing, to pull in a Rate.
4. Choose a Label Size: Small, Medium, Large, or one of the three "Label Only" sizes (see section 6.5).
5. Click Print. A running total above the grid always shows exactly how many labels are about to be sent before you commit.

Nothing on this page is ever saved as a document — each print is a one-time action.

6.1 My Printer

Each user can save their own default printer from the My Printer control at the top of the page — it shows a live list of whatever printers QZ Tray (or Zebra Browser Print) currently sees connected on that PC. This is remembered per user, not per browser or per site, and takes priority over the site-wide Default Label Printer set in Settings whenever it is chosen.

6.2 Scan Barcode

A dedicated Scan Barcode box above the item grid fills or updates a row the instant you scan or type into it - the same mechanism ERPNext itself uses everywhere else (e.g. Purchase Order's own "Scan Barcode" field). It accepts:

- A registered Item Barcode, a Serial No., a Batch No., or (if none of those match) a plain typed Item Code.
- An external USB/Bluetooth scanner, which types the value then usually sends Enter - but Enter is not required: the value commits automatically a short moment after typing/scanning stops.
- The small camera icon inside the box, which opens a live camera scan dialog for a device with no external scanner hardware (a phone or tablet, for example).

Scanning an item that already has a row behaves differently depending on how that item is tracked:

- Plain item (no batch/serial tracking): the existing row's No. of Barcodes increases by 1.
- Batch-tracked item: the match also checks the batch - the same item with a different batch gets its own new row instead.
- Serial-tracked item: the newly-scanned serial is added to that row's existing Serial Nos list, rather than bumping a count.

6.3 Load from Excel/CSV

Two buttons above the item grid handle bulk entry instead of adding rows one at a time:

- Download Template exports whatever rows are currently in the grid as a CSV file - a convenient starting point to edit in a spreadsheet and re-upload, or just a blank header row if the grid is empty.
- Load from File reads an uploaded .xlsx, .xls, or .csv file back into the grid. Each row is checked independently (the Item exists and is not disabled, any Barcode/Batch No./Serial No. given actually belongs to that Item) - a bad row is skipped and listed in a results dialog, never a whole-file failure.

> **Note:** A file with more than 1,000 rows is rejected outright with a clear message, rather than silently cut off at 1,000.

6.4 Stock Qty

A read-only Stock Qty column shows each row's on-hand quantity - purely informational, never printed on the label. It is left blank for service/non-stock Items, and is scoped to the row's own Batch No. when one is set, otherwise it reflects the Item's overall on-hand total.

6.5 "Label Only" sizes

For an Item that genuinely has no barcode, three extra Label Size choices - Small Label Only, Medium Label Only, Large Label Only - print a clean label with text only (Item Name, Item Code, UOM, Rate, per that size's own Display Rule) and no barcode field at all. Each reuses its own base size's physical dimensions (a Small Label Only prints on the exact same size label as Small).

> **Note:** This is a deliberate, explicit choice only. The regular Small/Medium/Large sizes are completely unaffected - they still require a real barcode and still show an error if the Item has none, exactly as before this feature existed. Choosing a real barcode is never bypassed automatically just because one happens to be missing.

6.6 Items with more than one barcode

If an Item has more than one barcode registered, the page asks which one to use, rather than silently guessing. You can mark one barcode per item as the Default in the Item master's Barcodes table (a Default checkbox, one per row) — when exactly one barcode is marked Default, it is pre-selected automatically and the prompt is skipped, while you can still pick a different barcode for that print if needed. Only one row per item may be marked Default at a time.

6.7 Batch No. and Serial No. printing

For items that track batches or serial numbers, the page also lets you pick exactly which Batch No., or which Serial No.(s), a label is for — provided Batch/Serial Printing in Settings (section 5.3) allows that type site-wide:

- Batch-tracked items: pick the Batch No. from a dropdown filtered to that item.
- Serial-tracked items: a dialog lists the available serial numbers for that item (and batch, if applicable) to choose one or more from.

- The chosen Batch No./Serial No. is encoded directly into the label's own barcode, so scanning that label anywhere in ERPNext resolves straight to that specific batch or unit — the same way scanning the item's regular barcode resolves to the item.
- Whether Batch No./Serial No. actually appears as text on the printed label (in addition to being encoded in the barcode itself) is controlled separately, per label size, by Display Rules described in section 5.4.

7. Printing from Purchase Order / Purchase Receipt / Purchase Invoice / Subcontracting Order

When Purchase Document Printing is enabled in Settings, a Print Barcode button appears on submitted documents of the selected type.

1. Open a submitted Purchase Order, Purchase Receipt, or Purchase Invoice (whichever is selected in Settings) and click Print Barcode.

2. The Print Barcode page opens pre-filled with that document's stock items — service items are left out automatically. Each line's own Batch No./Serial No. (both the legacy and the Serial and Batch Bundle recording style), Company, and Tax Template come straight from the document, not from any site or user default.

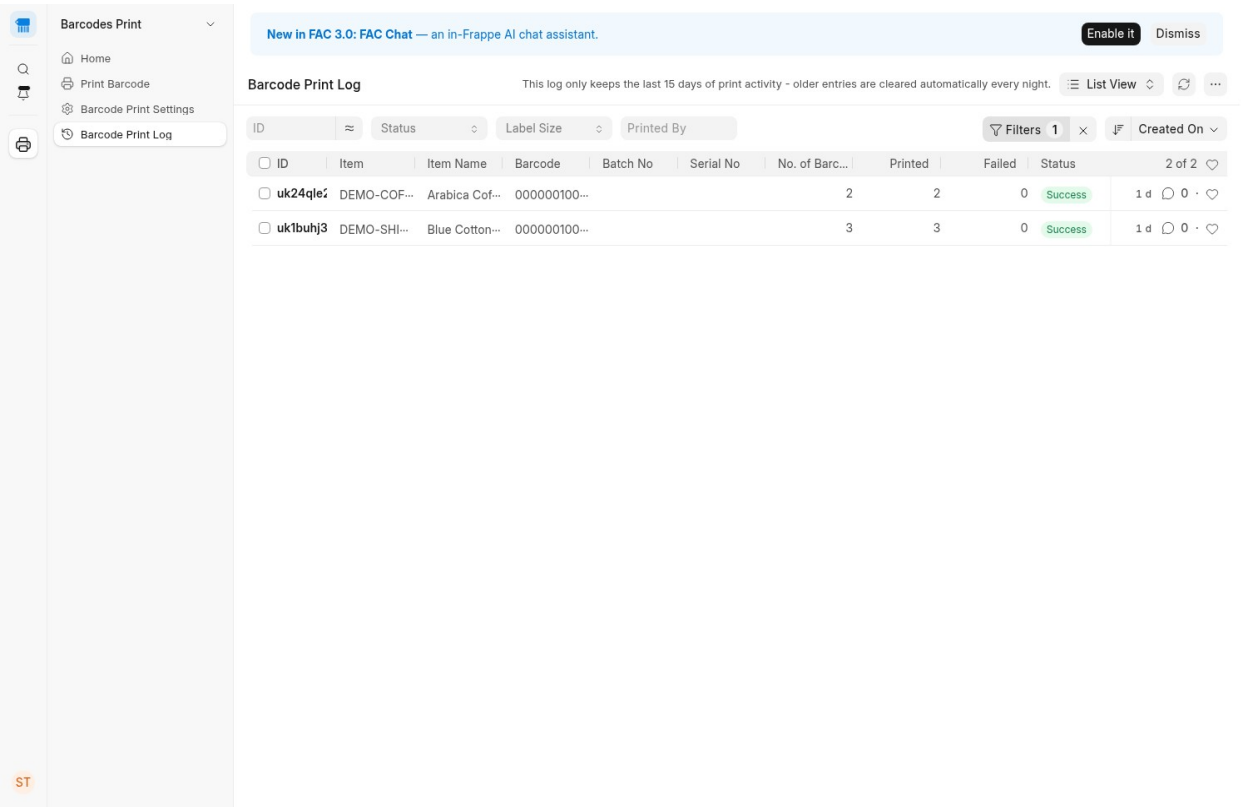
3. Each line's starting quantity defaults to whatever is still allowed: the line's own quantity, plus any configured extra allowance, minus what has already been printed for that line.

4. A running Barcodes Printed counter is kept per line, and re-checked on the server on every print — raising the request beyond what is allowed is rejected even if the page is edited.

Subcontracting Orders work the same way and are always available for this, regardless of which of the three document types is selected — only the order's finished-goods items are offered, never the supplied raw materials. A Return (credit note) Purchase Invoice is explicitly out of scope and never gets the button.

8. Barcode Print Log

Every print attempt is recorded automatically — item, barcode, quantity requested versus actually printed versus failed, a Success/Failed status with the failure reason on record, and which Purchase Order/Receipt line it came from, if any.

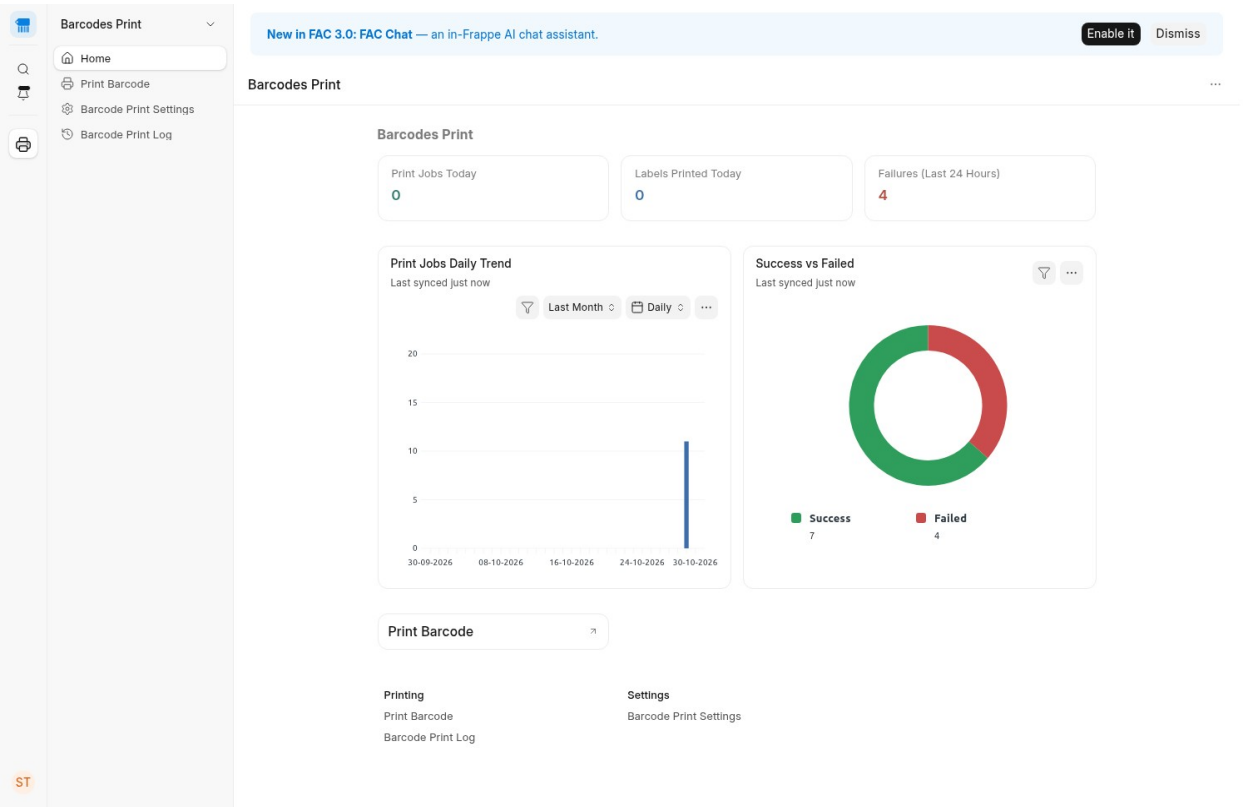


Note: QZ Tray and Zebra Browser Print each send one combined job per print action, so success/failure is tracked per print action, not per individual label within it.

Note: Log entries are kept for 15 days and swept automatically by a daily background job; they are an audit trail, not a long-term record.

9. Dashboard

The Barcodes Print workspace home screen shows a live dashboard, computed straight from the Barcode Print Log with no setup required: Number Cards for Print Jobs Today, Labels Printed Today, and Failures Last 24h, plus Success vs. Failed and Print Jobs Daily Trend charts.



10. Barcode Types Supported

EAN-13, EAN-8, UPC-A, UPC-E, Code 39, Code 128, ITF / ITF-14 / GTIN-14, Codabar, QR Code, and Data Matrix each get their own native printer barcode command. ISBN/ISSN/JAN/PZN and anything else without a dedicated symbology fall back to Code 128, which safely encodes any text or alphanumeric value.

If an Item's barcode has no type recorded, the type is auto-detected from the value itself, in this priority order: EAN-13, then EAN-8, then UPC, then Code 39, then Code 128. QR Code and Data Matrix are only ever used when explicitly set on the Item, and printing one is blocked outright if the chosen label size and display settings would render it too small to scan reliably.

11. Compatibility

Barcodes Print installs and runs on both Frappe/ERPNext v15 and v16, with CI running the full test suite on both versions on every push. On v15, the Desk sidebar entry and app icon are skipped automatically, since the underlying sidebar/icon doctypes are v16-only; every doctype, page, and printing feature works the same on both versions.

12. Support

For issues or feature requests, please use the project's GitHub repository.